

MI PANEL API

Version 2.15

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REVISION HISTORY

Revision No.	Description	Date
2.03	Initial release	04/12/2018
2.04	Added new API: MI_PANEL_SetSscConfig	06/06/2019
2.05	Added new mipi dsi config parameters: MI_U8 u8SyncCalibrate MI_U16 u16VirHsyncSt MI_U16 u16VirHsyncEnd MI_U16 u16VsyncRef MI_U16 u16DataClkSkew	08/09/2019
2.06	Added new mipi dsi config parameters: MI_U8 u8PolCh0 MI_U8 u8PolCh1 MI_U8 u8PolCh2 MI_U8 u8PolCh3 MI_U8 u8PolCh4	10/14/2019
2.07	Added output pixel format enum E_MI_PNL_OUTPUT_565BIT_MODE	12/09/2019
2.08	Redefined SSC parameter MI_PANEL_SscConfig_t	03/13/2020
2.09	Added new API: MI_PANEL_InitDev MI_PANEL_DeInitDev	04/26/2020
2.10	- Added interface type to API - Removed needless API	06/18/2020
2.11	Added BT656 interface enum	07/03/2020
2.12	Added new interface type	09/03/2020
2.13	- Declare that these functions are not supported - MI_PANEL_SetPowerOn MI_PANEL_GetPowerOn - MI_PANEL_SetBacklight - MI_PANEL_GetBacklight - MI_PANEL_SetBacklightLevel - MI_PANEL_GetBacklightLevel	06/25/2021
2.14	- Added new interface type - E_MI_PNL_INTF_BT1120_DDR	12/08/2021
2.15	Modify MI_PANEL_DeInit and MI_PANEL_DeinitDev input param	02/28/2022

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1. API LIST

The panel module provides the following APIs:

Name of API	Function
MI_PANEL_Init	Initialize panel module
MI_PANEL_DeInit	Deinitialize panel module
MI_PANEL_GetTotalNum	Get total number of panel parameters
MI_PANEL_SetPowerOn	Set panel power on
MI_PANEL_GetPowerOn	Get panel power status
MI_PANEL_SetBacklight	Set backlight enable
MI_PANEL_GetBacklight	Get backlight enable status
MI_PANEL_SetBacklightLevel	Set backlight brightness value
MI_PANEL_GetBacklightLevel	Get backlight brightness value
MI_PANEL_SetSscConfig	Set spread spectrum related configuration
MI_PANEL_SetMipiDsiConfig	Set MIPI related configuration
MI_PANEL_SetTimingConfig	Set Timing related configuration
MI_PANEL_SetDrvCurrentConfig	Set driving current configuration
MI_PANEL_SetOutputPattern	Set panel output pattern
MI_PANEL_SetPanelParam	Set panel information
MI_PANEL_GetPanelParam	Get current panel configuration information
MI_PANEL_GetTimingConfig	Get Timing related configuration information
MI_PANEL_InitDev	Initialize panel device
MI_PANEL_DeInitDev	De-initialize panel device

1.1. MI_PANEL_Init

➤ Function

Initialize panel module

➤ Syntax

```
MI_S32 MI_PANEL_Init(MI\_PANEL\_IntfType\_e eIntfType);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Example

N/A.

➤ Related API

[MI_PANEL_DeInit](#)

1.2. MI_PANEL_DeInit

➤ Function

Deinitialize panel module

➤ Syntax

```
MI_S32 MI_PANEL_DeInit(MI\_PANEL\_IntfType\_e eIntfType);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Example

N/A.

➤ Related API

[MI_PANEL_Init](#)

1.3. MI_PANEL_GetTotalNum

➤ Function

Get total number of panel parameters

➤ Syntax

```
MI_S32 MI_PANEL_GetTotalNum(MI_U32 *pu32TotalNum);
```

➤ Parameter

Parameter Name	Description	Input/Output
pu32TotalNum	Total number of panel parameters received by the pointer	Output

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Note

Not supported yet

➤ Example

N/A.

➤ Related API

N/A.

1.4. MI_PANEL_SetPowerOn

➤ Function

Set panel power on.

➤ Syntax

```
MI_S32 MI_PANEL_SetPowerOn(MI\_PANEL\_IntfType\_e eIntfType, MI\_PANEL\_PowerConfig\_t *pstPowerCfg);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstPowerCfg	Panel power enable status	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Note

- Before setting the panel power enable, initialize panel first.
- Currently not supported, it is recommended that the app directly control GPIO.

➤ Example

N/A.

➤ Related API

[MI_PANEL_GetPowerOn](#)

1.5. MI_PANEL_GetPowerOn

➤ Function

Get panel power enable status.

➤ Syntax

```
MI_S32 MI_PANEL_GetPowerOn(MI\_PANEL\_IntfType\_e eIntfType ,  
MI\_PANEL\_PowerConfig\_t *pstPowerCfg);
```

➤ Syntax

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstPowerCfg	Panel power enable status received by the pointer	Output

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Note

Currently not supported, it is recommended that the app directly control GPIO.

➤ Example

N/A.

➤ Related API

[MI_PANEL_SetPowerOn](#)

1.6. MI_PANEL_SetBacklight

➤ Function

Set panel backlight enable.

➤ Syntax

```
MI_S32 MI_PANEL_SetBacklight(MI\_PANEL\_IntfType\_e eIntfType ,
MI\_PANEL\_BackLightConfig\_t *pstBackLightCfg);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstBackLightCfg	Panel backlight enable status	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Note

Currently not supported, it is recommended that the app directly control PWM.

➤ Example

N/A.

➤ Related API

[MI_PANEL_GetBackLight](#)

1.7. MI_PANEL_GetBacklight

➤ Function

Get panel backlight enable status.

➤ Syntax

```
MI_S32 MI_PANEL_GetBacklight(MI\_PANEL\_IntfType\_e eIntfType ,
MI\_PANEL\_BackLightConfig\_t *pstBackLightCfg);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstBackLightCfg	Panel backlight enable status received by the pointer	Output

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Note

Currently not supported, it is recommended that the app directly control PWM.

➤ Example

N/A.

➤ Related API

[MI_PANEL_SetBackLight](#)

1.8. [MI_PANEL_SetBacklightLevel](#)

➤ Function

Set backlight brightness value.

➤ Syntax

```
MI_S32 MI_PANEL_SetBacklightLevel(MI\_PANEL\_IntfType\_e eIntfType ,
MI\_PANEL\_BackLightConfig\_t *pstBackLightCfg);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstBackLightCfg	Backlight brightness value	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h

- Library: libmi_panel.a / libmi_panel.so

➤ Note

- Before calling this function, set the panel backlight enable first.
- Currently not supported, it is recommended that the app directly control PWM.

➤ Example

N/A.

➤ Related API

[MI_PANEL_GetBacklightLevel](#)

1.9. MI_PANEL_GetBacklightLevel

➤ Function

Get panel backlight brightness value.

➤ Syntax

```
MI_S32 MI_PANEL_GetBacklightLevel(MI\_PANEL\_IntfType eIntfType ,  
MI\_PANEL\_BackLightConfig\_t *pstBackLightCfg);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstBackLightCfg	Panel backlight brightness value received by the pointer	Output

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Note

- Before calling this function, set the panel backlight enable first.
- Currently not supported, it is recommended that the app directly control PWM.

➤ Example

N/A.

➤ Related API

[MI_PANEL_SetBacklightLevel](#)

1.10. MI_PANEL_SetSscConfig

➤ Function

Set spread spectrum related configuration

➤ Syntax

```
MI_S32 MI_PANEL_SetSscConfig(MI\_PANEL\_IntfType\_e eIntfType , MI\_PANEL\_SscConfig\_t
*pstSscCfg);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstSscCfg	Spread spectrum related configuration	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Note

N/A.

➤ Example

N/A.

➤ Related API

N/A.

1.11. MI_PANEL_SetMipiDsiConfig

➤ Function

Set MIPI interface related configuration

➤ Syntax

```
MI_S32 MI_PANEL_SetMipiDsiConfig(MI\_PANEL\_IntfType\_e eIntfType ,
MI\_PANEL\_MipiDsiConfig\_t *pstMipiDsiCfg);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstMipiDsiCfg	MIPI interface related configuration	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Note

This function is for use with MIPI panel only. To use this function, [MI_PANEL_SetPanelParam](#) must be called as well.

➤ Example

N/A.

➤ Related API

N/A.

1.12. MI_PANEL_SetTimingConfig

➤ Function

Set Timing related configuration

➤ Syntax

```
MI_S32 MI_PANEL_SetTimingConfig(MI\_PANEL\_IntfType\_e eIntfType ,  
MI\_PANEL\_TimingConfig\_t *pstTimingCfg);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstTimingCfg	Timing related configuration	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Note

N/A.

➤ Example

N/A.

➤ Related API

[MI_PANEL_GetTimingConfig](#)

1.13. MI_PANEL_SetDrvCurrentConfig

➤ Function

Set driving capability related configuration

➤ Syntax

```
MI_S32 MI_PANEL_SetDrvCurrentConfig(MI\_PANEL\_IntfType\_e eIntfType ,  
MI\_PANEL\_DrvCurrentConfig\_t *pstDrvCurrentCfg);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstDrvCurrentCfg	Driving capability related configuration	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Note

N/A.

➤ Example

N/A.

➤ Related API

N/A.

1.14. MI_PANEL_SetOutputPattern

➤ Function

Set output pattern

➤ Syntax

```
MI_S32 MI_PANEL_SetOutputPattern(MI\_PANEL\_IntfType\_e eIntfType ,  
MI\_PANEL\_TestPatternConfig\_t *pstTestPatternCfg);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstTestPatternCfg	Output pattern related configuration	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Note

For debugging (against LVDS at the present)

➤ Example

N/A.

➤ Related API

N/A.

1.15. MI_PANEL_SetPanelParam

➤ Function

Set panel information

➤ Syntax

```
MI_S32 MI_PANEL_SetPanelParam(MI\_PANEL\_IntfType\_e eIntfType ,  
MI\_PANEL\_ParamConfig\_t *pstParamCfg);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstParamCfg	Panel parameter information	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Note

N/A.

➤ Example

N/A.

➤ Related API

[MI_PANEL_GetPanelParam](#)

1.16. MI_PANEL_GetPanelParam

➤ Function

Get panel parameter

➤ Syntax

```
MI_S32 MI_PANEL_GetPanelParam(MI\_PANEL\_IntfType\_e eIntfType ,  
MI\_PANEL\_ParamConfig\_t *pstParamCfg);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstParamCfg	Panel parameter information	Output

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Note

[MI_PANEL_SetPanelParam](#) must be called first

➤ Example

N/A.

➤ Related API

[MI_PANEL_SetPanelParam](#)

1.17. MI_PANEL_GetTimingConfig

➤ Function

Get Timing related configuration

➤ Syntax

```
MI_S32 MI_PANEL_GetTimingConfig(MI\_PANEL\_IntfType\_e eIntfType ,  
MI\_PANEL\_TimingConfig\_t *pstTimingCfg);
```

➤ Parameter

Parameter Name	Description	Input/Output
eIntfType	PANEL type	Input
pstTimingCfg	Timing related configuration	Output

- Return Value
 - Zero: Successful
 - Non-zero: Failed, see [error code for](#) details
- Requirement
 - Header: mi_panel.h, mi_panel_datatype.h
 - Library: libmi_panel.a / libmi_panel.so
- Note

N/A.
- Example

N/A.
- Related API

[MI_PANEL_SetTimingConfig](#)

1.18. MI_PANEL_InitDev

- Function

Initialize panel module
- Syntax


```
MI_S32 MI_PANEL_InitDev(MI\_PANEL\_InitParam\_t *pstInitParam);
```
- Parameter

Parameter Name	Description	Input/Output
pstInitParam	Initialization Parameter	Input
- Return Value
 - Zero: Successful
 - Non-zero: Failed, see [error code for](#) details
- Requirement
 - Header: mi_panel.h, mi_panel_datatype.h
 - Library: libmi_panel.a / libmi_panel.so
- Example

N/A.
- ※ Note
 - For Version 2.09 and above, we recommend using this interface to replace the original [MI_PANEL_Init](#) interface.
- Related API

[MI_PANEL_DeInitDev](#)

1.19. MI_PANEL_DeInitDev

➤ Function

Deinitialize panel module

➤ Syntax

```
MI_S32 MI_PANEL_DeInitDev(MI\_PANEL\_InitParam\_t *pstInitParam);
```

➤ Parameter

Parameter Name	Description	Input/Output
pstInitParam	Initialization Parameter	Input

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [error code for](#) details

➤ Requirement

- Header: mi_panel.h, mi_panel_datatype.h
- Library: libmi_panel.a / libmi_panel.so

➤ Example

N/A.

※ Note

- For Version 2.09 and above, we recommend using this interface to replace the original [MI_PANEL_DeInit](#) interface.

➤ Related API

[MI_PANEL_InitDev](#)

2. PANEL DATA TYPE

The table below lists the PANEL related data types:

MI_PANEL_PowerConfig_t	Define panel power enable status structure
MI_PANEL_BackLightConfig_t	Define panel backlight structure
MI_PANEL_SscConfig_t	Define panel spread spectrum configuration structure
MI_PANEL_MipiDsiConfig_t	Define MIPI interface configuration structure
MI_PANEL_MipiDsiLaneMode_e	Define MIPI interface Lane mode enumeration
MI_PANEL_MipiDsiFormat_e	Define MIPI interface Format enumeration
MI_PANEL_MipiDsiCtrlMode_e	Define MIPI interface Ctrl Mode enumeration
MI_PANEL_TimingConfig_t	Define panel Timing information structure
MI_PANEL_DrvCurrentConfig_t	Define driving capability configuration structure
MI_PANEL_TestPatternConfig_t	Define output pattern information structure
MI_PANEL_ParamConfig_t	Define panel parameter information structure
MI_PANEL_IntfType_e	Define Link type enumeration
MI_PANEL_AspectRatio_e	Define Aspect Ratio enumeration
MI_PANEL_TiBitMode_e	Define TiBit Mode enumeration
MI_PANEL_OutputFormatBitMode_e	Define Output Format Bit Mode enumeration
MI_PANEL_OutputTimingMode_e	Define panel output timing change mode enumeration
MI_PANEL_InitParam_t	Define the initialized parameter of PANEL device

NOTE: This section includes most of the key data types. For data types not listed here, please refer to `mi_panel_datatype.h`.

2.1. MI_PANEL_PowerConfig_t

- Description
Define panel power enable status structure
- Definition


```
typedef struct
{
    MI_BOOL bEn;
}MI_PANEL_PowerConfig_t;
```
- Related Data Type and Interface
 - [MI_PANEL_SetPowerOn](#)
 - [MI_PANEL_GetPowerOn](#)

2.2. MI_PANEL_BackLightConfig_t

- Description
Define panel backlight information structure
- Definition


```
typedef struct
{
    MI_BOOL bEn;
    MI_U16 u16Level;
}MI_PANEL_BackLightConfig_t;
```
- Member

Member	Description
bEn	Enable panel backlight or not
u16Level	Set backlight level if bEn = TRUE
- Related Data Type and Interface
 - [MI_PANEL_SetBacklight](#)
 - [MI_PANEL_GetBacklight](#)
 - [MI_PANEL_SetBacklightLevel](#)
 - [MI_PANEL_GetBacklightLevel](#)

2.3. MI_PANEL_SscConfig_t

- Description
Define panel spread spectrum configuration structure

➤ Definition

```
typedef struct
{
    MI_BOOL bEn;
    MI_U16 u16Freq;
    MI_U16 u16Ratio;
}MI_PANEL_SscConfig_t;
```

➤ Member

Member	Description
bEn	Panel spread spectrum function enable status
u16Freq	Modulation rate
u16Ratio	Modulation depth

➤ Note

u16Freq should be greater than 15KHz.

➤ Related Data Type and Interface

[MI_PANEL_SetSscConfig](#)

2.4. MI_PANEL_MipiDsiConfig_t

➤ Description

Define MIPI interface configuration structure

➤ Definition

```
typedef struct
{
    MI_U8 u8HsTrail;
    MI_U8 u8HsPrpr;
    MI_U8 u8HsZero;
    MI_U8 u8ClkHsPrpr;
    MI_U8 u8ClkHsExit;
    MI_U8 u8ClkTrail;
    MI_U8 u8ClkZero;
    MI_U8 u8ClkHsPost;
    MI_U8 u8DaHsExit;
    MI_U8 u8ContDet;

    MI_U8 u8Lpx;
    MI_U8 u8TaGet;
    MI_U8 u8TaSure;
    MI_U8 u8TaGo;
```

```

MI_U16 u16Hactive;
MI_U16 u16Hpw;
MI_U16 u16Hbp;
MI_U16 u16Hfp;
MI_U16 u16Vactive;
MI_U16 u16Vpw;
MI_U16 u16Vbp;
MI_U16 u16Vfp;

```

```

MI_U16 u16BlIp;
MI_U16 u16Fps;

```

```

MI_PANEL_MipiDsiLaneMode_e enLaneNum;
MI_PANEL_MipiDsiFormat_e enformat;
MI_PANEL_MipiDsiCtrlMode_e enCtrl;

```

```

MI_U8 *pu8CmdBuf;
MI_U32 u32CmdBufSize;

```

```

MI_U8 u8SyncCalibrate;
MI_U16 u16VirHsyncSt;
MI_U16 u16VirHsyncEnd;
MI_U16 u16VsyncRef;
MI_U16 u16DataClkSkew;

```

```

MI_U8 u8PolCh0;
MI_U8 u8PolCh1;
MI_U8 u8PolCh2;
MI_U8 u8PolCh3;
MI_U8 u8PolCh4;

```

```

}MI_PANEL_MipiDsiConfig_t;

```

➤ Related Data Type and Interface

[MI_PANEL_MipiDsiLaneMode_e](#)

[MI_PANEL_MipiDsiFormat_e](#)

[MI_PANEL_MipiDsiCtrlMode_e](#)

[MI_PANEL_SetMipiDsiConfig](#)

2.5. MI_PANEL_MipiDsiLaneMode_e

➤ Description

Define MIPI interface Lane mode enumeration

➤ Definition

```
typedef enum
{
    E_MI_PNL_MIPI_DSI_LANE_NONE    = 0,
    E_MI_PNL_MIPI_DSI_LANE_1      = 1,
    E_MI_PNL_MIPI_DSI_LANE_2      = 2,
    E_MI_PNL_MIPI_DSI_LANE_3      = 3,
    E_MI_PNL_MIPI_DSI_LANE_4      = 4,
} MI_PANEL_MipiDsiLaneMode_e;
```

➤ Related Data Type and Interface

[MI_PANEL_MipiDsiConfig_t](#)

[MI_PANEL_SetMipiDsiConfig](#)

2.6. MI_PANEL_MipiDsiFormat_e

➤ Description

Define MIPI interface Format enumeration

➤ Definition

```
typedef enum
{
    E_MI_PNL_MIPI_DSI_RGB565      = 0,
    E_MI_PNL_MIPI_DSI_RGB666      = 1,
    E_MI_PNL_MIPI_DSI_LOOSELY_RGB666 = 2,
    E_MI_PNL_MIPI_DSI_RGB888      = 3,
} MI_PANEL_MipiDsiFormat_e;
```

➤ Related Data Type and Interface

[MI_PANEL_MipiDsiConfig_t](#)

[MI_PANEL_SetMipiDsiConfig](#)

2.7. MI_PANEL_MipiDsiCtrlMode_e

➤ Description

Define MIPI interface Ctrl Mode enumeration

➤ Definition

```
typedef enum
{
    E_MI_PNL_MIPI_DSI_CTRL_CMD_MODE = 0,
    E_MI_PNL_MIPI_DSI_CTRL_SYNC_PULSE = 1,
    E_MI_PNL_MIPI_DSI_CTRL_SYNC_EVENT = 2,
    E_MI_PNL_MIPI_DSI_BURST_MODE = 3,
} MI_PANEL_MipiDsiCtrlMode_e;
```

- Related Data Type and Interface

[MI_PANEL_MipiDsiConfig_t](#)

[MI_PANEL_SetMipiDsiConfig](#)

2.8. MI_PANEL_TimingConfig_t

- Description

Define panel Timing information structure

- Definition

```
typedef struct
{
    MI_U16 u16HSyncWidth;
    MI_U16 u16HSyncBackPorch;
    MI_U16 u16HSyncFrontPorch;

    MI_U16 u16VSyncWidth;
    MI_U16 u16VSyncBackPorch;
    MI_U16 u16VSyncFrontPorch;

    MI_U16 u16HStart;
    MI_U16 u16VStart;

    MI_U16 u16Height;
    MI_U16 u16Width;

    MI_U16 u16Htotal;
    MI_U16 u16Vtotal;

    MI_U16 u16Dclk;
}MI_PANEL_TimingConfig_t;
```

- Related Data Type and Interface

[MI_PANEL_SetTimingConfig](#)

[MI_PANEL_GetTimingConfig](#)

2.9. MI_PANEL_DrvCurrentConfig_t

- Description

Define driving capability configuration structure

➤ Definition

```
typedef struct
{
    MI_U16 u16DrvCurrent;
}MI_PANEL_DrvCurrentConfig_t;
```

➤ Related Data Type and Interface

[MI_PANEL_SetDrvCurrentConfig](#)

2.10. MI_PANEL_TestPatternConfig_t

➤ Description

Define output pattern structure

➤ Definition

```
typedef struct
{
    MI_BOOL bEn;
    MI_U16 u16R;
    MI_U16 u16G;
    MI_U16 u16B;
}MI_PANEL_TestPatternConfig_t;
```

➤ Related Data Type and Interface

[MI_PANEL_SetOutputPattern](#)

2.11. MI_PANEL_ParamConfig_t

➤ Description

Define panel parameter structure

➤ Definition

```
typedef struct
{
    const char *pPanelName;    ///< PanelName
    MI_U8 u8Dither;            ///< PANEL_DITHER, keep the setting
    MI_PANEL_IntfType_e eIntfType;    ///< PANEL_LINK

    //////////////////////////////////////
    // Board related setting
    //////////////////////////////////////
    MI_U8 u8DualPort;          ///< VOP_21[8], MOD_4A[1], PANEL_DUAL_PORT, refer to
u8DoubleClick
    MI_U8 u8SwapPort;          ///< MOD_4A[0], PANEL_SWAP_PORT, refer to "LVDS output
```

app note" A/B channel swap

```
MI_U8 u8SwapOdd_ML;    ///< PANEL_SWAP_ODD_ML
MI_U8 u8SwapEven_ML;   ///< PANEL_SWAP_EVEN_ML
MI_U8 u8SwapOdd_RB;    ///< PANEL_SWAP_ODD_RB
MI_U8 u8SwapEven_RB;   ///< PANEL_SWAP_EVEN_RB
```

MI_U8 u8SwapLVDS_POL; ///< MOD_40[5], PANEL_SWAP_LVDS_POL, for differential P/N swap

```
MI_U8 u8SwapLVDS_CH;    ///< MOD_40[6], PANEL_SWAP_LVDS_CH, for pair swap
MI_U8 u8PDP10BIT;       ///< MOD_40[3], PANEL_PDP_10BIT, for pair swap
MI_U8 u8LVDS_TI_MODE;    ///< MOD_40[2], PANEL_LVDS_TI_MODE, refer to "LVDS output app note"
```

////////////////////////////////////

// For TTL Only

////////////////////////////////////

```
MI_U8 u8DCLKDelay;      ///< PANEL_DCLK_DELAY
MI_U8 u8InvDCLK;        ///< MOD_4A[4], PANEL_INV_DCLK
MI_U8 u8InvDE;          ///< MOD_4A[2], PANEL_INV_DE
MI_U8 u8InvHSync;       ///< MOD_4A[12], PANEL_INV_HSYNC
MI_U8 u8InvVSync;       ///< MOD_4A[3], PANEL_INV_VSYNC
```

////////////////////////////////////

// Output driving current setting

////////////////////////////////////

// driving current setting (0x00=4mA, 0x01=6mA, 0x02=8mA, 0x03=12mA)

```
MI_U8 u8DCLKCurrent;    ///< define PANEL_DCLK_CURRENT
MI_U8 u8DECurrent;      ///< define PANEL_DE_CURRENT
MI_U8 u8ODDDataCurrent;  ///< define PANEL_ODD_DATA_CURRENT
MI_U8 u8EvenDataCurrent; ///< define PANEL_EVEN_DATA_CURRENT
```

////////////////////////////////////

// panel on/off timing

////////////////////////////////////

```
MI_U16 u16OnTiming1;    ///< time between panel & data while turn on power
MI_U16 u16OnTiming2;    ///< time between data & back light while turn on power
MI_U16 u16OffTiming1;   ///< time between back light & data while turn off power
MI_U16 u16OffTiming2;   ///< time between data & panel while turn off power
```

////////////////////////////////////

// panel timing spec.

////////////////////////////////////

// sync related

```
MI_U16 u16HSyncWidth;   ///< VOP_01[7:0], PANEL_HSYNC_WIDTH
```

```

MI_U16 u16HSyncBackPorch;    ///< PANEL_HSYNC_BACK_PORCH, no register setting,
provide value for query only
MI_U16 u16VSyncWidth;        ///< define PANEL_VSYNC_WIDTH
MI_U16 u16VSyncBackPorch;    ///< define PANEL_VSYNC_BACK_PORCH

// DE related
MI_U16 u16HStart;            ///< VOP_04[11:0], PANEL_HSTART, DE H Start
(PANEL_HSYNC_WIDTH + PANEL_HSYNC_BACK_PORCH)
MI_U16 u16VStart;            ///< VOP_06[11:0], PANEL_VSTART, DE V Start
MI_U16 u16Width;             ///< PANEL_WIDTH, DE width (VOP_05[11:0] = HEnd = HStart
+ Width - 1)
MI_U16 u16Height;            ///< PANEL_HEIGHT, DE height (VOP_07[11:0], = Vend =
VStart + Height - 1)

// DClk related
MI_U16 u16MaxHTotal;         ///< PANEL_MAX_HTOTAL. Reserved for future using.
MI_U16 u16HTotal;            ///< VOP_0C[11:0], PANEL_HTOTAL
MI_U16 u16MinHTotal;         ///< PANEL_MIN_HTOTAL. Reserved for future using.
MI_U16 u16MaxVTotal;         ///< PANEL_MAX_VTOTAL. Reserved for future using.
MI_U16 u16VTotal;            ///< VOP_0D[11:0], PANEL_VTOTAL
MI_U16 u16MinVTotal;         ///< PANEL_MIN_VTOTAL. Reserved for future using.
MI_U16 u16MaxDCLK;           ///< PANEL_MAX_DCLK. Reserved for future using.
MI_U16 u16DCLK;              ///< LPLL_0F[23:0], PANEL_DCLK
MI_U16 u16MinDCLK;           ///< PANEL_MIN_DCLK. Reserved for future using.

                                ///< spread spectrum
MI_U16 u16SpreadSpectrumStep; ///< move to board define, no use now.
MI_U16 u16SpreadSpectrumSpan; ///< move to board define, no use now.

MI_U8 u8DimmingCtl;          ///< Initial Dimming Value
MI_U8 u8MaxPWMVal;           ///< Max Dimming Value
MI_U8 u8MinPWMVal;           ///< Min Dimming Value

MI_U8 u8DeinterMode;         ///< define PANEL_DEINTER_MODE, no use now
MI_PANEL_AspectRatio_e ePanelAspectRatio; ///< Panel Aspect Ratio, provide information
to upper layer application for aspect ratio setting.

MI_U16 u16LVDSTxSwapValue;
MI_PANEL_TiBitMode_e eTiBitMode;           ///< MOD_4B[1:0], refer to "LVDS
output app note"
MI_PANEL_OutputFormatBitMode_e eOutputFormatBitMode;

MI_U8 u8SwapOdd_RG;          ///< define PANEL_SWAP_ODD_RG
MI_U8 u8SwapEven_RG;         ///< define PANEL_SWAP_EVEN_RG

```

```

MI_U8 u8SwapOdd_GB;      ///< define PANEL_SWAP_ODD_GB
MI_U8 u8SwapEven_GB;     ///< define PANEL_SWAP_EVEN_GB

/**
 * Others
 */
MI_U8 u8DoubleClk;        ///< LPLL_03[7], define Double Clock,LVDS dual mode
MI_U32 u32MaxSET;         ///< define PANEL_MAX_SET
MI_U32 u32MinSET;         ///< define PANEL_MIN_SET
MI_PANEL_OutputTimingMode_e eOutTimingMode; ///

```

➤ Related Data Type and Interface

[MI_PANEL_IntfType_e](#)

[MI_PANEL_AspectRatio_e](#)

[MI_PANEL_TiBitMode_e](#)

[MI_PANEL_OutputFormatBitMode_e](#)

[MI_PANEL_OutputTimingMode_e](#)

[MI_PANEL_GetPanelParam](#)

2.12. MI_PANEL_IntfType_e

➤ Description

Define panel Interface Type enumeration

➤ Definition

```

typedef enum
{
    E_MI_PNL_INTF_TTL,          ///< TTL type
    E_MI_PNL_INTF_LVDS,         ///< LVDS type
    E_MI_PNL_INTF_RSDS,         ///< RSDS type
    E_MI_PNL_INTF_MINILVDS,     ///< TCON
    E_MI_PNL_INTF_ANALOG_MINILVDS, ///< Analog TCON
    E_MI_PNL_INTF_DIGITAL_MINILVDS, ///< Digital TCON
    E_MI_PNL_INTF_MFC,          ///< Ursa (TTL output to Ursa)
    E_MI_PNL_INTF_DAC_I,        ///< DAC output
    E_MI_PNL_INTF_DAC_P,        ///< DAC output
    E_MI_PNL_INTF_PDPLVDS,      ///< For PDP(Vsync use Manually MODE)
    E_MI_PNL_INTF_EXT,          ///< EXT LPLL TYPE
    E_MI_PNL_INTF_MIPI_DSI,     ///< Mipi DSI
    E_MI_PNL_INTF_BT656,        ///< BT656 interface
    E_MI_PNL_INTF_BT601,        ///< BT601

```

```

        E_MI_PNL_INTF_BT1120,          ///< BT1120
        E_MI_PNL_INTF_MCU_TYPE,        ///< MCU Type
        E_MI_PNL_INTF_SRGB,            ///< sRGB
        E_MI_PNL_INTF_TTL_SPI_IF,      ///< TTL with SPI init interface
        E_MI_PNL_INTF_BT1120_DDR,
    }MI_PANEL_IntfType_e;

```

- Note
 - E_MI_PNL_INTF_BT1120 is single edge sampling; E_MI_PNL_INTF_BT1120_DDR is double edge sampling
- Related Data Type and Interface
 - [MI_PANEL_ParamConfig_t](#)
 - [MI_PANEL_GetPanelParam](#)

2.13. MI_PANEL_AspectRatio_e

- Description
 - Define panel Aspect Ratio enumeration
- Definition


```

typedef enum
{
    E_MI_PNL_ASPECT_RATIO_4_3    = 0,  ///< set aspect ratio to 4 : 3
    E_MI_PNL_ASPECT_RATIO_WIDE,    ///< set aspect ratio to 16 : 9
    E_MI_PNL_ASPECT_RATIO_OTHER,    ///< reserved for other aspect ratio other than
    4:3/ 16:9
}MI_PANEL_AspectRatio_e;

```
- Related Data Type and Interface
 - [MI_PANEL_ParamConfig_t](#)
 - [MI_PANEL_GetPanelParam](#)

2.14. MI_PANEL_TiBitMode_e

- Description
 - Define panel TiBit Mode enumeration
- Definition


```

typedef enum
{
    E_MI_PNL_TI_10BIT_MODE = 0,
    E_MI_PNL_TI_8BIT_MODE  = 2,
    E_MI_PNL_TI_6BIT_MODE  = 3,
} MI_PANEL_TiBitMode_e;

```

- Related Data Type and Interface

[MI_PANEL_ParamConfig_t](#)

[MI_PANEL_GetPanelParam](#)

2.15. MI_PANEL_OutputFormatBitMode_e

- Description

Define panel output format Bit Mode enumeration

- Definition

```
typedef enum
{
    E_MI_PNL_OUTPUT_10BIT_MODE = 0, //default is 10bit, because 8bit panel can use 10bit
    config and 8bit config.
    E_MI_PNL_OUTPUT_6BIT_MODE = 1, //but 10bit panel(like PDP panel) can only use 10bit
    config.
    E_MI_PNL_OUTPUT_8BIT_MODE = 2, //and some PDA panel is 6bit.
    E_MI_PNL_OUTPUT_565BIT_MODE = 3,
} MI_PANEL_OutputFormatBitMode_e;
```

- Related Data Type and Interface

[MI_PANEL_ParamConfig_t](#)

[MI_PANEL_GetPanelParam](#)

2.16. MI_PANEL_OutputTimingMode_e

- Description

Define panel output timing change mode

- Definition

```
typedef enum
{
    E_MI_PNL_CHG_DCLK = 0,    ///

```

- Related Data Type and Interface

[MI_PANEL_ParamConfig_t](#)

[MI_PANEL_GetPanelParam](#)

2.17. MI_PANEL_InitParam_t

- Description

Define panel device initialization parameter

➤ Definition

```
typedef struct MI_PANEL_InitParam_s
{
    MI_PANEL_IntfType_e eIntfType; ///PANEL Interface Type enumerate .
} MI_PANEL_InitParam_t;
```

➤ Related Data Type and Interface

[MI_PANEL_InitDev](#)

3. PANEL ERROR CODE

Error Code	Macro Definition	Description
0xA00B2011	MI_ERR_PANEL_INVALID_PARAM	Invalid parameter
0xA00B2006	MI_ERR_PANEL_NULL_PTR	Null pointer
0xA00B2001	MI_ERR_PANEL_INVALID_DEVID	Invalid device ID
0xA00B2018	MI_ERR_PANEL_DRV_FAILED	PANEL driving failed
0xA00B2015	MI_ERR_PANEL_NOT_INIT	PANEL not initialized
0xA00B2008	MI_ERR_PANEL_NOT_SUPPORT	Unsupported operation